Karatina University

Department of Computing & Informatics

COURSE OUTLINE
B.Sc(CS) & B.Sc(IT) Third Year Semester One
Unit Code: COM 332 Unit Title: Advanced Database Systems

Lecturer: Dr. Malanga N. Kennedy **Email:** kmalanga@karu.ac.ke **Tel:** 0735742905

Contact Hours: 45
Prerequisites:

Course purpose
The aim of this course is to encourage the usage of advanced database management systems for effective data management.

Learning outcomes
By the end of the course, the learner should be able to:

i. Describe Physical and Logical Database Architectures,
ii. Describe and use Security and Access Control,
iii. Describe and Set-up Disaster and Recovery Processes,
iv. Describe Replication Mechanisms and Uses,
v. Describe Database Monitoring and Performance Tuning.

Course outline

Week	Topic	Sub-topic	Remarks
1.	DBMS architectures	• Multi-user database architectures, • Network and distributed databases, • Client-server databases, • Multimedia databases.	
	Database Modeling	• ERMs, OOP, relational	
2.	Query Languages	• SQL, QBE, relational algebra	
3.	Database Creation and Management	• DDL Commands, DML Commands, Views, and Reports.	
4.	Database Optimization	• Hashing and Indexing, • Query evaluation, • Query optimization techniques.	
5.	**C.A.T 1**		
6.	Transaction processing concepts	• Introduction to transaction processing. • Transactions and system	

		concepts.	
7.	Transaction processing concepts	• Desirable properties of transactions. • Characterizing Schedules Based on recoverability.	
8.	Transaction processing concepts	• Characterizing Schedules Based on serializability. • Transaction support in SQL.	
9.	Concurrency control techniques	• Two-phase Locking Techniques for Concurrency Control. • Concurrency Control Based on Timestamp Ordering.	
10	**C.A.T 2**		
11.	Concurrency control techniques	• Multi-version Concurrency Control Techniques. • Concurrency Control issues.	
12.	Database recovery techniques	• Recovery Techniques. • Recovery Techniques based on Deferred Update. • Recovery Techniques based on immediate Update. • Shadow Paging. • The ARIES Recovery Algorithms	
13.	Trends in database technology	• Data mining, • Data warehousing	
14.	Case study	• MySQL case study	
15 & 16		• End semester examinations	

Teaching methodology
Lectures, practical demonstrations, and hands-on laboratory sessions and projects.

Instructional materials
Projectors, white board markers, white boards, internet.

Course assessment

Type	**Weighting (%)**
Cats, assignments and practical	30%
Final written Examination	70 %
Total	**100 %**

Course texts books

1. Elmasri, R., & Navathe, S. (2015). *Fundamentals of database systems* (Vol. 7). Pearson. ISBN: 978-0133970777
2. Sciore.E (2020) Database Design and Implementation: Second Edition (Data-Centric Systems and Applications)-m I SBN-13: 978-3030338350
3. Alex P (2019)Database Internals: A Deep Dive into How Distributed Data Systems Work 1st Edition - ISBN-13: 978-1492040347

Reference text books

1. Inmon.B & Puppini.F (2020)The Unified Star Schema: An Agile and Resilient Approach to Data Warehouse and Analytics Design Paperback – October 12, 2020
2. Jukic.N and Vrbsky.S (2016) Database Systems: Introduction to Databases and Data

Course journals

1. Advances in data Analysis and Classification ISSN1 1862-5347
2. Journal of computer science and technology ISSN- 0916-534276
3. Annals Of software Engineering ISSN 1022-7091Vrbsky.S (2016) Database Systems: Introduction to Databases and Data

Reference journals

1. Journal of computer science and Technology ISSN 1000-9000
2. Journal of Science and Technology ISSN 1860-4749
3. Directory of Open Access (DOAJ): Computer Science, ISSN 0034-567352